

Symbiosis Institute of Technology, Nagpur

DEPARTMENT OF COMPUTER SCIENCE • JAVASCRIPT PROGRAMMING LAB

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Course: Semester V (B.Tech)

Experiment No. 2

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File Path: EXP2/index.html , EXP2/script.js

Experiment Title

Online Shopping Bill Calculator

Software / Tools Required

1. Visual Studio Code / Antigravity IDE
2. Google Chrome (or modern web browser)
3. HTML5
4. CSS3
5. JavaScript (ES6)

Experiment Program Code

Task 2.a — Product Entry Interface and Output Bill Table

File: EXP2/index.html

The HTML document renders an online shopping bill calculator with input controls for Product Name, Unit Price (₹), and Quantity, an action button to compute total amounts, and a summary breakdown table.

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Shopping Bill Calculator</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: Segoe UI, Arial, sans-serif;
    }
    body {
      background: linear-gradient(135deg, #0f172a, #1d4ed8, #38bdf8);
      min-height: 100vh;
      display: flex;
      justify-content: center;
      align-items: center;
      padding: 30px;
    }
  </style>
</head>
</html>
```

```
.container {
  width: 700px;
  background: #fff;
  border-radius: 20px;
  padding: 35px;
  box-shadow: 0 15px 35px rgba(0,0,0,.3);
}
h1 {
  text-align: center;
  color: #0f172a;
}
p {
  text-align: center;
  color: gray;
  margin-bottom: 25px;
}
label {
  display: block;
  margin-top: 15px;
  margin-bottom: 8px;
  font-weight: bold;
  color: #1e293b;
}
input {
  width: 100%;
  padding: 14px;
  border: 1px solid #cbd5e1;
  border-radius: 10px;
  font-size: 16px;
}
input:focus {
  outline: none;
  border-color: #2563eb;
  box-shadow: 0 0 8px rgba(37,99,235,.4);
}
button {
  width: 100%;
  padding: 15px;
  margin-top: 25px;
  background: #2563eb;
  color: white;
  border: none;
  border-radius: 10px;
  font-size: 18px;
  cursor: pointer;
}
button:hover {
  background: #1d4ed8;
}
table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 30px;
}
table, th, td {
  border: 1px solid #d1d5db;
}
th {
  background: #2563eb;
  color: white;
  padding: 12px;
}
td {
  padding: 12px;
  text-align: center;
}
tr:nth-child(even) {
  background: #f8fafc;
}
.footer {
  margin-top: 35px;
  text-align: center;
```

```

        border-top: 2px solid #ddd;
        padding-top: 20px;
        color: #555;
    }
    .footer b {
        color: #1d4ed8;
    }
</style>
</head>
<body>
<div class="container">
    <h1>Online Shopping Bill Calculator</h1>
    <p>Enter Product Details</p>

    <label>Product Name</label>
    <input type="text" id="product" placeholder="Enter Product Name">

    <label>Price (₹)</label>
    <input type="number" id="price" placeholder="Enter Price">

    <label>Quantity</label>
    <input type="number" id="quantity" placeholder="Enter Quantity">

    <button onclick="calculateBill()">Calculate Bill</button>

    <table id="billTable">
        <tr>
            <th>Product</th>
            <th>Price</th>
            <th>Quantity</th>
            <th>Total Bill</th>
        </tr>
        <tr>
            <td id="pName">-</td>
            <td id="pPrice">-</td>
            <td id="pQty">-</td>
            <td id="totalBill">-</td>
        </tr>
    </table>

    <div class="footer">
        <p><b>Developed By</b></p>
        <p><b>Tanmay Sankulwar</b></p>
        <p>PRN : <b>24070521058</b></p>
    </div>
</div>
<script src="script.js"></script>
</body>
</html>

```

Task 2.b — Numerical Parsing and Bill Calculation Logic

File: EXP2/script.js

The JavaScript script retrieves user inputs, uses `parseFloat()` and `parseInt()` to parse currency and unit quantities, validates the numbers using `isNaN()`, computes the total bill (`price * quantity`), formats the currency to two decimal places with `toFixed(2)`, and updates the table cells.

```

function calculateBill(){
    var product=document.getElementById("product").value;
    var price=parseFloat(document.getElementById("price").value);
    var quantity=parseInt(document.getElementById("quantity").value);

    if(product=="" || isNaN(price) || isNaN(quantity)){
        alert("Please enter all details.");
        return;
    }
}

```

```
}  
  
var total=price*quantity;  
  
document.getElementById("pName").innerHTML=product;  
document.getElementById("pPrice").innerHTML="₹"+price.toFixed(2);  
document.getElementById("pQty").innerHTML=quantity;  
document.getElementById("totalBill").innerHTML="₹"+total.toFixed(2);  
}
```

Output

- Initial State:** The card displays empty input fields and default hyphens (-) across Product, Price, Quantity, and Total Bill table columns.
- Validation Check:** If the product name is blank or either price or quantity is not a valid number (`isNaN`), the function generates an alert box: "Please enter all details."
- Calculation & Formatted Display:** Upon valid input (e.g. Product: Laptop, Price: 55000, Quantity: 2), the total is computed as ₹110000.00 and immediately rendered into the bill table with formatted Rupee symbols.

Screenshot

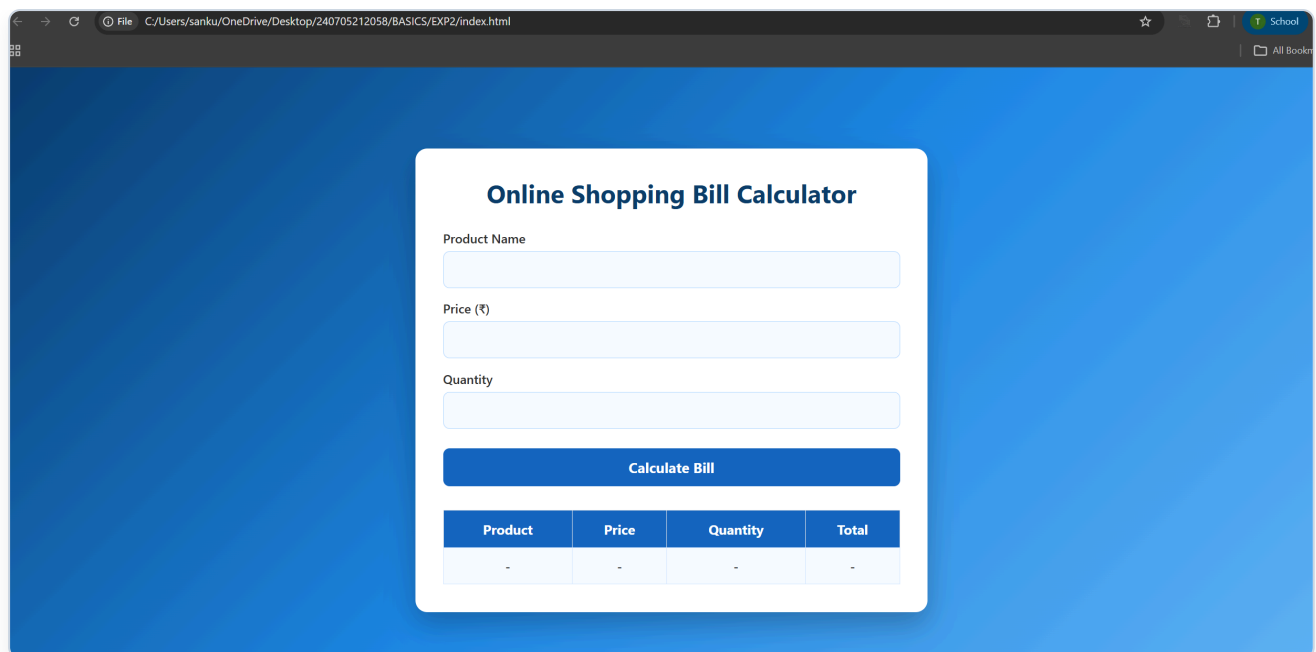
Initial Input & Validation State

The screenshot shows a web browser window with the URL `C:/Users/sanku/OneDrive/Desktop/24070521058_Javascript/Exp2/index.html`. The page features a dark blue header and a white central card titled "Online Shopping Bill Calculator" with the subtitle "Enter Product Details". Inside the card, there are three input fields labeled "Product Name", "Price (₹)", and "Quantity", each with a placeholder text "Enter Product Name", "Enter Price", and "Enter Quantity" respectively. Below these fields is a blue "Calculate Bill" button. At the bottom of the card, it says "Developed By Tanmay Sankulwar" and "PRN : 24070521058".

Product	Price	Quantity	Total Bill
-	-	-	-

Shopping Bill Calculator Interface

Computed Bill Output



The screenshot shows a web browser window with the address bar displaying "C:/Users/sanku/OneDrive/Desktop/24070521058/BASICS/EXP2/index.html". The page has a blue gradient background. In the center, there is a white card titled "Online Shopping Bill Calculator". Inside the card, there are three input fields labeled "Product Name", "Price (₹)", and "Quantity". Below these fields is a blue button labeled "Calculate Bill". At the bottom of the card, there is a table with four columns: "Product", "Price", "Quantity", and "Total". The table currently shows dashes "-" in all four columns.

Product	Price	Quantity	Total
-	-	-	-

Shopping Bill Calculation Result

Result / Conclusion

Experiment 2 successfully demonstrated arithmetic operations, data type conversion, and DOM updates in JavaScript. Concepts reinforced include parsing floating-point and integer numbers (`parseFloat` , `parseInt`), validation with `isNaN()` , numerical formatting with `.toFixed(2)` , and dynamically binding calculated financial values to HTML table elements.